

Detecting AI-Generated Veterinary Radiographs: A Dual-Stream Approach Against Pet Insurance Fraud

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Abstract—Generative models can now synthesize medical images realistic enough to deceive both human experts and automated systems, a risk recently confirmed for human radiographs by a 2026 RSNA study in which board-certified radiologists misclassified roughly one in five AI-generated chest X-rays as authentic. This vulnerability directly threatens insurance underwriting, where fraudulent diagnostic imagery can be used to inflate or fabricate claims. While medical deepfake detection has received growing attention in human radiology (e.g., MedForensics, M3DSynth), no prior published dataset or method addresses *veterinary* diagnostic imaging, despite documented and costly fraud in the pet insurance industry, estimated at 3–7 loss-ratio points (USD 300K–700K per USD 10M in premium). This paper introduces (1) a legally re-distributable dataset of canine thoracic radiographs paired with AI-generated synthetic counterparts, built by re-licensing an existing CC BY 4.0 Mendeley Data collection and augmenting it with a trained generative adversarial network, and (2) a dual-stream detector combining spatial CNN features (ResNet-18) with a radially-averaged frequency-domain descriptor designed to expose the periodic upsampling artifacts characteristic of GAN-based generation. We release the resulting dataset publicly on Kaggle to seed further research in this unexplored niche. On a held-out test set, our detector reaches 100% accuracy, precision, recall, and AUC; we discuss why this ceiling result reflects the limited visual realism of a DCGAN trained on only 152 images rather than a solved problem, and outline the harder cross-generator (GAN vs. diffusion) evaluation needed to validate real-world robustness.

Index Terms—veterinary radiology, medical deepfake detection, generative adversarial networks, insurance fraud, frequency-domain forensics, synthetic medical imaging

RESUMEN (SPANISH ABSTRACT)

Los modelos generativos ya pueden sintetizar imágenes médicas suficientemente realistas como para engañar tanto a expertos humanos como a sistemas automáticos, un riesgo confirmado en 2026 por un estudio de la RSNA en el que radiólogos certificados clasificaron erróneamente como auténticas cerca de una de cada cinco radiografías de tórax generadas por IA. Esta vulnerabilidad amenaza directamente la suscripción de pólizas de seguros, donde una imagen diagnóstica fraudulenta puede usarse para inflar o fabricar un reclamo. Mientras la detección de deepfakes médicos ha recibido atención creciente en radiología humana (MedForensics, M3DSynth), ningún dataset ni método publicado aborda la imagen diagnóstica *veterinaria*, pese a que el fraude en seguros de mascotas es un problema documentado y costoso

para la industria (entre 3 y 7 puntos de loss ratio, 300 mil a 700 mil dólares por cada 10 millones en primas). Este artículo presenta un dataset legalmente redistribuible de radiografías caninas de tórax junto con contrapartes sintéticas generadas por IA, construido re-licenciando una colección CC BY 4.0 de Mendeley Data y aumentándola con una red generativa adversarial entrenada, y un detector de dos ramas que combina características espaciales (ResNet-18) con un descriptor de frecuencia que expone los artefactos periódicos del proceso de upsampling. El detector alcanza 100

I. INTRODUCTION

Generative adversarial networks (GANs) and diffusion models have reached a level of visual fidelity that makes synthetic medical images increasingly difficult to distinguish from real acquisitions. Tordjman et al., in a multi-institutional study published in *Radiology* in March 2026, exposed 17 radiologists from 12 institutions across six countries to a balanced set of 264 authentic and AI-generated chest radiographs (produced with tools including ChatGPT-based generators and Stanford’s RoentGen model); reader accuracy ranged from 58% to 92%, meaning a meaningful fraction of fabricated images escaped detection entirely, and the same vulnerability extended to automated multimodal models [1]. The study explicitly names insurance fraud, litigation, and factitious disorder (Munchausen syndrome) as realistic misuse scenarios [1].

This risk is not hypothetical for the pet insurance industry. Unlike human healthcare systems, veterinary medical records do not travel between clinics: there is no shared infrastructure to verify an animal’s diagnostic history across providers. Industry analyses estimate that claims fraud costs pet insurance managing general agents (MGAs) 3–7 points of loss ratio, translating to USD 300K–700K per USD 10M in written premium, while noting that even basic rules-based detection programs deliver a 3–10x return on investment [2]. As image-editing and generative tools become trivially accessible, fabricated or altered diagnostic imagery (a forged fracture, an exaggerated lesion) is a natural next step for claims fraud, exactly mirroring the human-healthcare threat already demonstrated by Tordjman et al. [1].

Despite this, a review of the medical deepfake detection literature reveals that every existing dataset and method targets

human imaging modalities. MedForensics, presented at MIC-CAI 2025, spans six human modalities (ultrasound, endoscopy, histopathology, MRI, CT, X-ray) and twelve generative models, explicitly measuring cross-generator generalization [3]. M3DSynth addresses AI-generated local manipulations in human 3D medical volumes [4]. Shahzad et al., comparing GAN- and diffusion-generated fakes directly, report the same generalization gap using human skin-cancer imagery [5]. None of this work, nor any dataset we could locate on Kaggle or elsewhere, considers veterinary diagnostic images.

This paper makes three contributions:

- 1) A legally clear, publicly re-distributed dataset of canine thoracic radiographs paired with GAN-generated synthetic counterparts, built by re-licensing an existing CC BY 4.0 dataset and republishing it on Kaggle with correct attribution to the original authors.
- 2) A dual-stream detector architecture that fuses spatial CNN features with a frequency-domain descriptor, adapted from the human-imaging forensics literature to a new, data-scarce veterinary domain.
- 3) An honest characterization of what a small, real, publicly available veterinary imaging dataset can and cannot support, intended as a reproducible starting point rather than a production-ready fraud detection system.

The remainder of this paper is organized as follows. Section II reviews related work on medical deepfake generation and detection. Section III describes the dataset and its licensing chain. Section IV presents the generation and detection methodology. Section V describes the experimental setup. Section VI reports results. Section VII discusses limitations and future work, and Section VIII concludes.

II. RELATED WORK

A. Synthetic medical image generation

GANs and diffusion models are both used to synthesize medical images across modalities including MRI, CT, X-ray, ultrasound, and dermoscopy, motivated primarily by data augmentation and privacy-preserving data sharing [6]. Diffusion models have recently overtaken GANs in perceptual image quality, including for chest X-ray generation, but their clinical reliability and the artifacts they leave behind remain comparatively underexplored relative to GANs [6].

B. Detecting AI-generated medical images

MedForensics is, to our knowledge, the most comprehensive published benchmark for medical deepfake detection, spanning six modalities and twelve generative models, and proposes DSKI, a dual-stage knowledge-infusing detector built on a vision-language feature space that outperforms prior methods and human experts in a visual Turing test [3]. Its architecture uses three parallel CNN streams targeting visual perception, resampling artifacts, and local inconsistency. M3DSynth focuses on localized manipulations within 3D volumes rather than whole-image synthesis [4]. Shahzad et al. train detectors on skin-cancer imagery against both GAN (DCGAN, Pix2Pix) and diffusion (DDPM, VQ-Diffusion) generators, confirming

that detectors calibrated to one generator family degrade sharply against the other, and propose domain-specific batch normalization to mitigate this distribution shift [5].

C. Frequency-domain forensics

Outside the medical domain, a substantial body of work shows that CNN- and GAN-based generators leave periodic, checkerboard-like artifacts in the frequency domain, a direct consequence of the transposed-convolution upsampling operations used to grow a low-resolution latent representation into a full-resolution image [7]. These artifacts are frequently invisible in the spatial domain but become a strong, generalizable signal once an image is transformed via FFT or DCT, and have been used as an auxiliary detection stream in several natural-image forensics pipelines [7]. To our knowledge, this signal has not previously been evaluated on veterinary radiographic imagery, where anatomical structures (rib spacing, cassette-induced grid patterns) already have their own periodic regularities that could plausibly interact with generator artifacts in either direction.

D. The veterinary gap

Existing machine-learning work on canine and feline radiographs, largely from Tahghighi et al., is limited to image-quality classification [8] and hemithorax symmetry assessment [9], both performed on private institutional archives rather than public data, and neither addresses synthetic image generation or detection. Public veterinary imaging datasets are themselves rare: most published work draws on closed PACS (picture archiving and communication system) archives from single institutions due to case-volume and business-confidentiality constraints, rather than open repositories comparable to those available for human radiology.

III. DATASET

A. Source data and licensing

The real-image component of our dataset originates from “Radiographic Dataset for VHS determination learning process,” collected by Flores Dueñas, Gaxiola Camacho, and Montañó Gómez at the Small Animal Veterinary Teaching Hospital, Veterinary Sciences Research Institute, Autonomous University of Baja California, and published on Mendeley Data under a CC BY 4.0 license (DOI: 10.17632/ktx4cj55pn.1) [10]. The collection comprises 152 latero-lateral thoracic radiographs from 156 canine patients, acquired in 2019–2020 using a Universal AV Choice X-Ray System with a Rayence XMARU 1417PGA-PCA digital detector, converted from DICOM to anonymized PNG format. We verified the license programmatically via the Mendeley Data public API and confirm it permits redistribution with attribution, which we provide here and in the republished Kaggle listing.

B. Synthetic counterpart generation

We trained a DCGAN-style generator directly on the 152 real radiographs (Section IV) to produce a matched set of synthetic “fake” radiographs. Because the source collection is

small, we treat this generator as a data-scarce, reproducible baseline rather than a state-of-the-art image synthesizer; Section VII discusses fine-tuned diffusion models as a natural extension.

C. Public re-release

We repackage the real images (with attribution) and our generated synthetic images into a single Kaggle dataset, `vet-radiographs-real-vs-ai-generated`, to our knowledge the first publicly available real-vs-synthetic veterinary radiograph collection. Kaggle’s terms explicitly permit re-hosting openly licensed third-party data with correct attribution, which we provide via the original DOI and author names in the dataset card.

IV. METHODOLOGY

A. Synthetic radiograph generator

We use a DCGAN [11] architecture: a five-layer transposed-convolution generator mapping a 100-dimensional latent vector to a 128×128 grayscale image, and a mirrored five-layer strided-convolution discriminator, both with batch normalization and, respectively, ReLU and LeakyReLU activations. Given the small training set, we apply aggressive on-the-fly augmentation (random horizontal flip, small affine jitter) and one-sided label smoothing (real label 0.9) to stabilize adversarial training.

B. Dual-stream detector

Following the architectural intuition behind MedForensics [3] and the frequency-forensics literature [7], our detector fuses two complementary signals rather than relying on spatial appearance alone:

- **Spatial stream:** a ResNet-18 backbone, pretrained on ImageNet and fine-tuned end-to-end, producing a 512-dimensional feature vector from the grayscale radiograph (replicated to three channels).
- **Frequency stream:** a radially-averaged log-power spectrum of the image’s 2D FFT, binned into 32 concentric annuli and normalized to $[0, 1]$, passed through a small two-layer MLP producing a 32-dimensional feature vector. This descriptor is designed to surface the periodic upsampling artifacts described in Section II regardless of their exact spatial location.

The two feature vectors are concatenated (544 dimensions) and passed through a two-layer classification head with dropout ($p=0.3$) to produce a single real/fake logit, trained with binary cross-entropy.

V. EXPERIMENTAL SETUP

All experiments were run on Apple silicon (M4 Pro) using PyTorch’s Metal Performance Shaders (MPS) backend; no CUDA hardware was used. The generator was trained for 800 epochs on the 152 real images (batch size 16, Adam, $\text{lr} = 2 \times 10^{-4}$), producing a pool of 2,300 synthetic radiographs, of which 456 (three times the real count) were sampled for detector training to bound class imbalance while the full

pool is released on Kaggle for reuse. Images were resized to 128×128 for both the generator and the detector. The combined dataset (152 real + 456 synthetic images, 608 total) was split 64/16/20 into train/validation/test (388/98/122) with stratified sampling on the real/fake label, using a fixed random seed (42) for reproducibility. The detector was trained with Adam ($\text{lr} = 1 \times 10^{-4}$) for 25 epochs, selecting the checkpoint with the highest validation AUC for final test-set evaluation. Code, exact hyperparameters, and the trained generator/detector weights accompany this paper in the project repository.

VI. RESULTS

On the held-out test set (122 images: 30 real, 92 synthetic), the dual-stream detector reaches 100% accuracy, precision, recall, F1, and AUC (Table I). Validation AUC reached 1.0 by epoch 1 and training loss converged to near zero within 20 epochs, indicating the classification task was fully separable for this generator.

TABLE I
TEST-SET PERFORMANCE OF THE DUAL-STREAM DETECTOR

Metric	Value
Accuracy	1.000
Precision	1.000
Recall	1.000
F1-score	1.000
AUC	1.000

We interpret this ceiling result cautiously rather than as evidence of a solved problem: a DCGAN trained on only 152 images at 128×128 resolution leaves generation artifacts (Figure 2) that are far more separable than those of a modern diffusion model would be. Section VII details why we treat this as a lower bound on task difficulty, not an upper bound on detector capability.

Fig. 1. ROC curve of the dual-stream detector on the held-out test set. The curve saturates at AUC = 1.0, consistent with the full separability discussed in the text.

Fig. 2. Average log-power spectrum of real radiographs (left), generated radiographs (center), and their difference (right), computed over the full 152 real and 2,300 generated images. The periodic structure visible in the difference saturates map corresponds to the transposed-convolution upsampling artifacts described in Section II.

VII. DISCUSSION AND LIMITATIONS

Dataset scale. 152 real radiographs is small relative to human-imaging forensics benchmarks such as MedForensics, and results here should be read as a reproducible proof of concept rather than a production-ready fraud detection system. The scarcity itself is evidence for the gap this paper documents: unlike human radiology, veterinary imaging has no equivalent of large public archives (e.g., NIH ChestX-ray14,

MIMIC-CXR), because most institutional PACS archives remain closed.

Interpreting the 100% result. A perfect test-set score is a warning sign as much as a result. With only 152 training images, our DCGAN converges to a generator whose outputs remain visually and spectrally distinguishable from real radiographs at essentially every frequency band, so the detection task is closer to distinguishing two different image distributions than to catching a near-miss forgery. We report it transparently rather than omit it, but we explicitly do not claim this generalizes to stronger generators. The correct reading is: our dual-stream architecture is a sound baseline, and the interesting open question, does it still separate real from fake once the generator is a fine-tuned diffusion model rather than a small DCGAN, remains unanswered and is the paper’s most direct future-work item.

GAN vs. diffusion generalization. Our generator is a DCGAN, not a diffusion model. The human-imaging literature has already shown that detectors trained against GAN artifacts generalize poorly to diffusion-generated images and vice versa [5]. Fine-tuning a diffusion model (e.g., via LoRA or DreamBooth) on this same 152-image collection, and testing whether our frequency-domain stream transfers across generator families, is the most direct next step and is left as future work.

Scope of imaging modality and species. The source dataset covers a single view (latero-lateral thoracic) in a single species (canine). Extending this pipeline to feline imaging, abdominal or orthopedic radiographs, and ultrasound would materially increase its relevance to real-world claims review, where fraud is not confined to one view or species.

Intended use. This work is intended as a research contribution and an openly available dataset to seed further study, not as a deployed fraud-detection product. Any operational use in claims adjudication would require substantially larger, prospectively validated data and regulatory/ethical review appropriate to insurance decision-making.

VIII. CONCLUSION

We identified a real, unaddressed gap between two active but previously disconnected lines of work: the growing literature on detecting AI-generated human medical images, and the well-documented, costly problem of veterinary insurance claims fraud. We built the first publicly available dataset pairing real and AI-generated veterinary radiographs, by legally re-licensing an existing CC BY 4.0 collection and augmenting it with a trained generator, and proposed a dual-stream detector that combines spatial and frequency-domain evidence. We release both the dataset and the full training pipeline to support future work in this niche, including the open question of cross-generator generalization between GANs and diffusion models in veterinary imaging.

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